

Universidade do Minho
Escola de Engenharia
Departamento de Electrónica Industrial

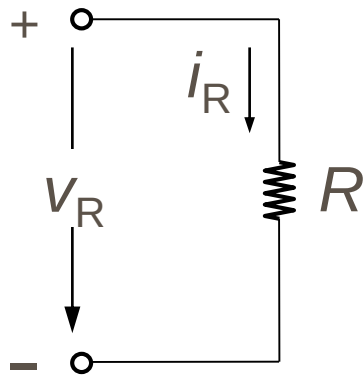
Engenharia de Telecomunicações e Informática

Eletrónica Geral

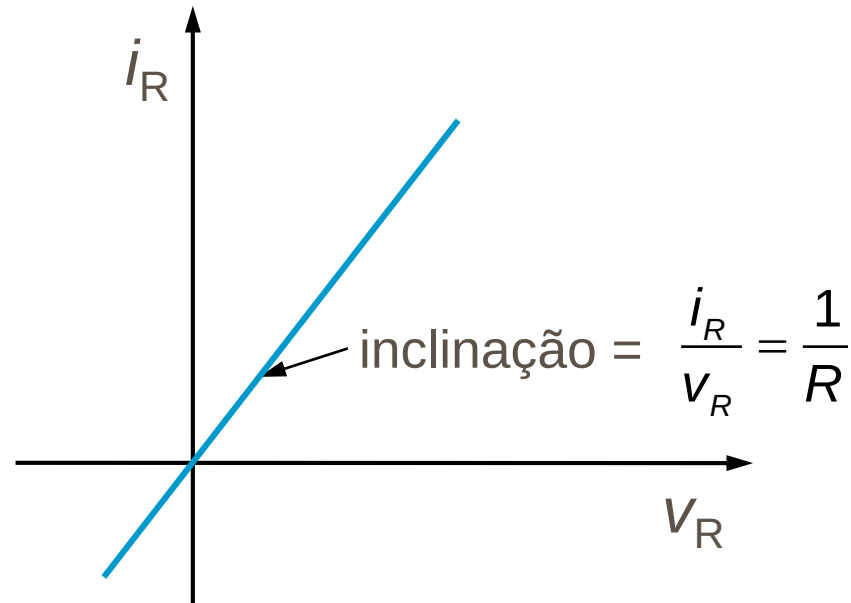
Departamento de Eletrónica Industrial

■ Comportamento do Díodo Ideal

■ Característica v - i de uma resistência



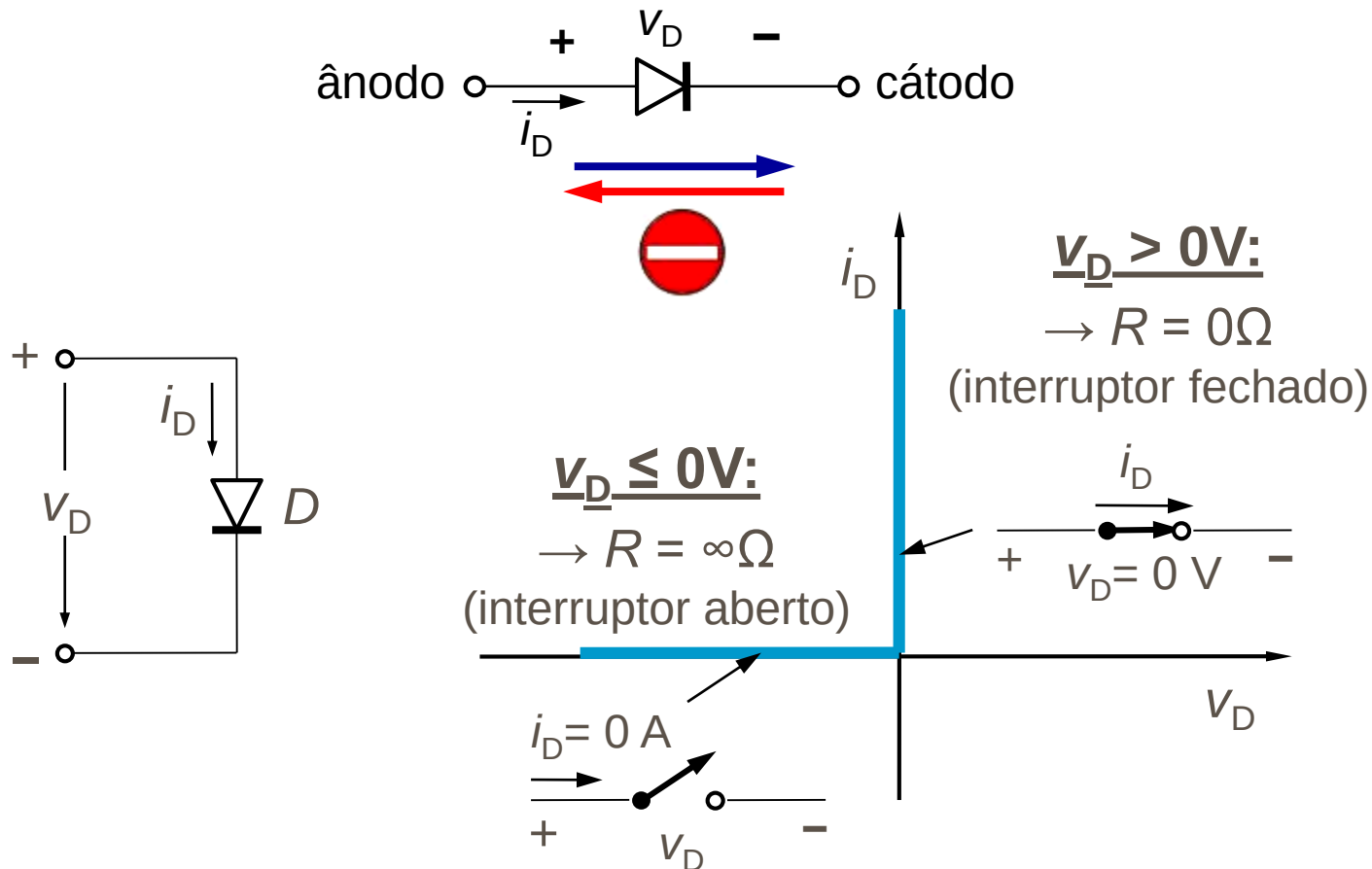
$$i_R = \frac{1}{R} \cdot V_R$$



Díodo Semicondutor

■ Comportamento do Díodo Ideal

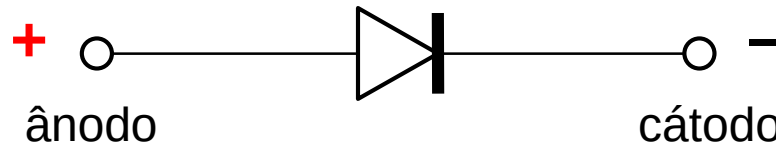
■ Característica v - i do díodo



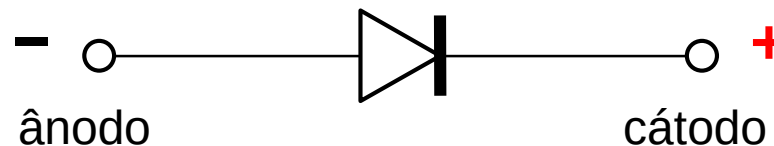
■ Comportamento do Díodo Ideal

■ Característica $v-i$ do díodo

Directamente polarizado



Inversamente polarizado



■ Comportamento do Díodo Ideal

- Análise de circuitos com díodos (exemplo)

$$v_i = V_m \text{sen}(\omega t)$$

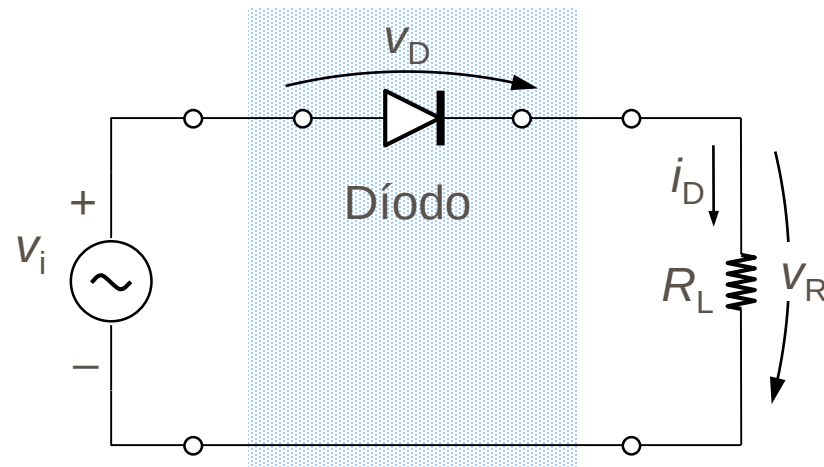
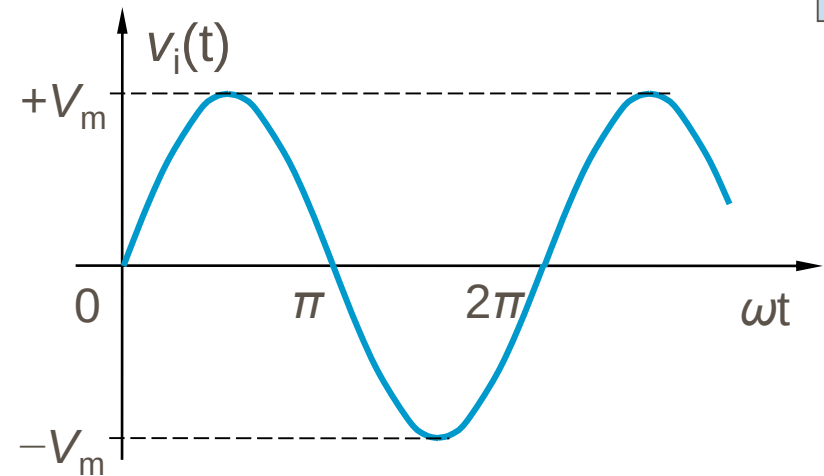
$$i_D = ?$$

$$v_i(t) = v_R + v_D$$

↓ ↓ ?

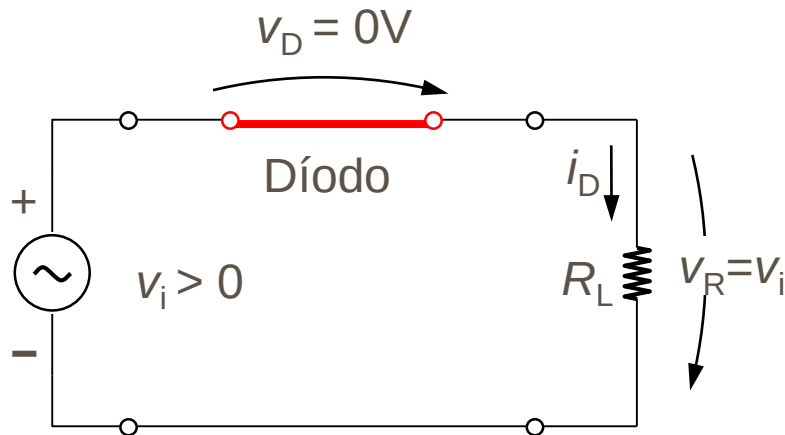
$$v_i(t) = R_L i_D + v_D$$

$$i_D = \frac{v_i - v_D}{R_L}$$



■ Comportamento do Díodo Ideal

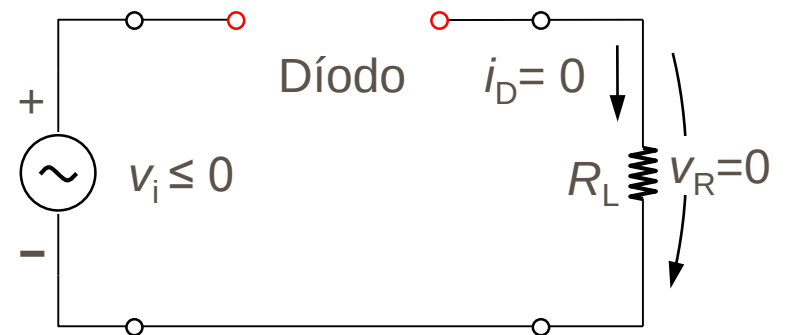
■ Exemplo



Para $v_i > 0$:

$$i_D = \frac{v_i}{R_L} = \frac{V_m}{R_L} \text{sen}(\omega t)$$

$$v_R = R_L i_D$$

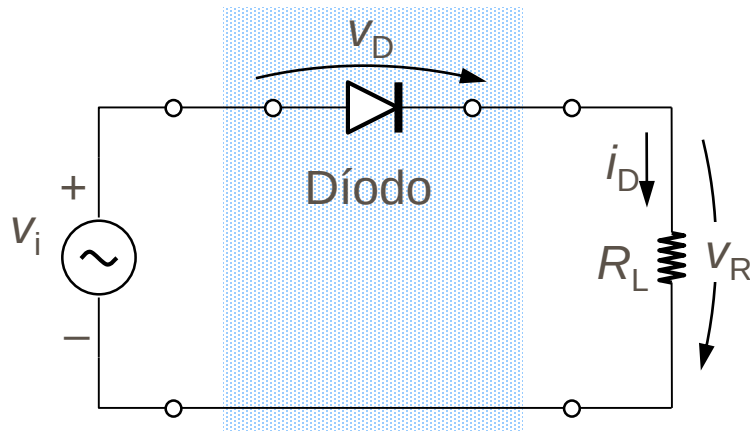


Para $v_i \leq 0$:

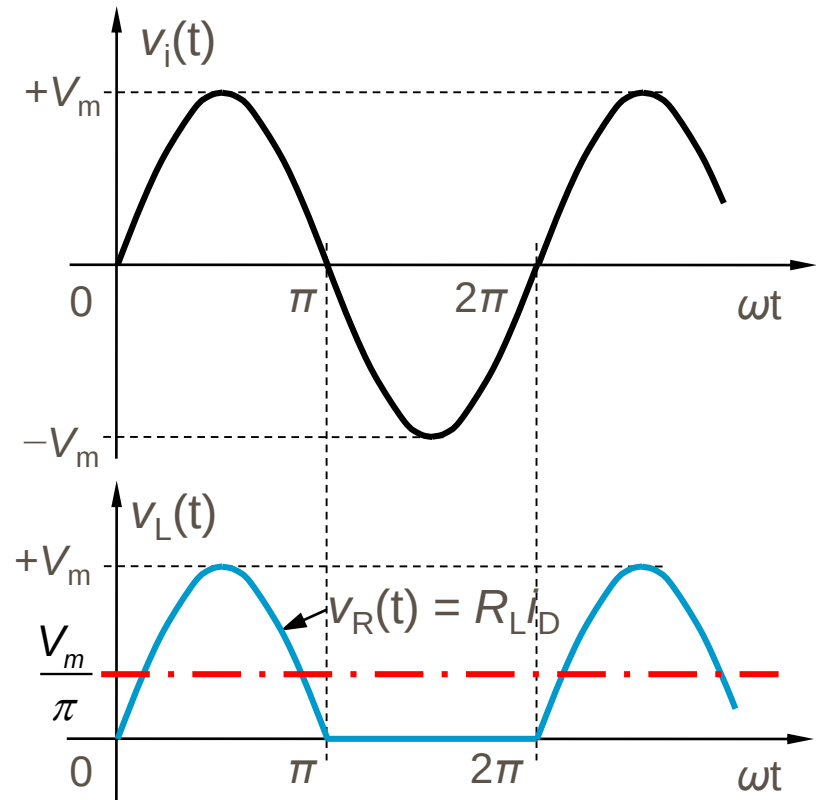
$$i_D = 0$$

■ Comportamento do Díodo Ideal

■ Exemplo

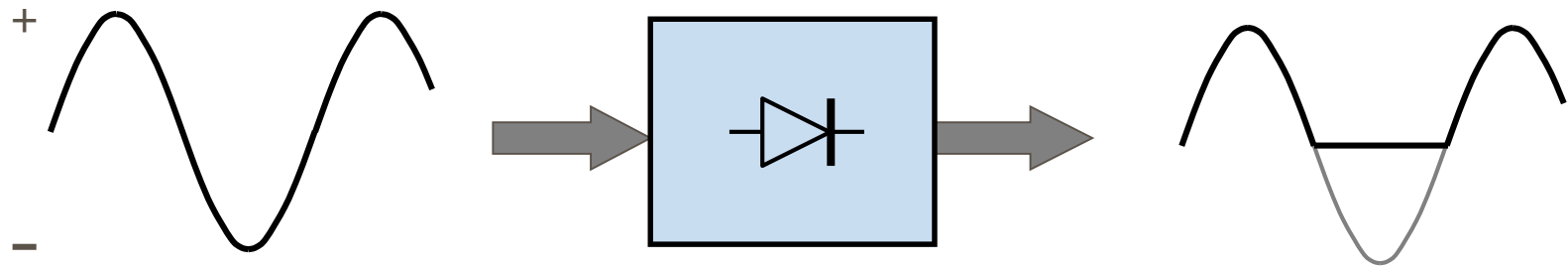


$$V_{Rmed} = \frac{1}{T} \int_0^T V_m \sin(\omega t) dt = \frac{V_m}{\pi}$$



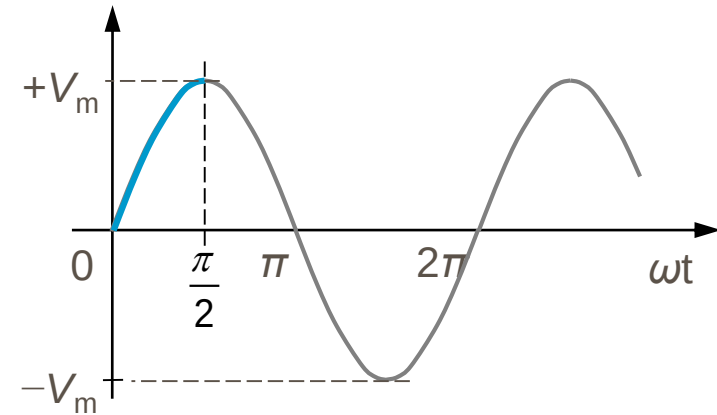
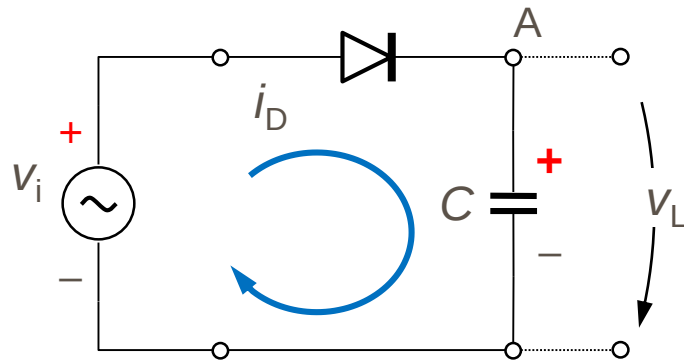
■ Comportamento do Díodo Ideal

■ Bloco Rectificador



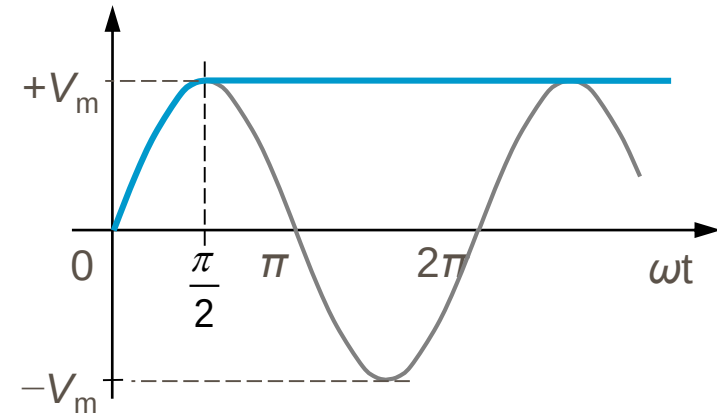
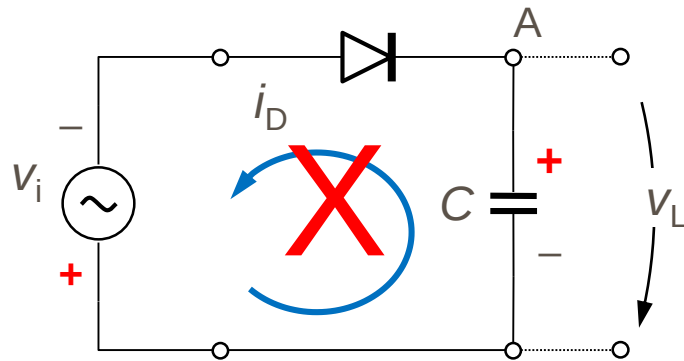
■ Comportamento do Díodo Ideal

■ Detector de Pico



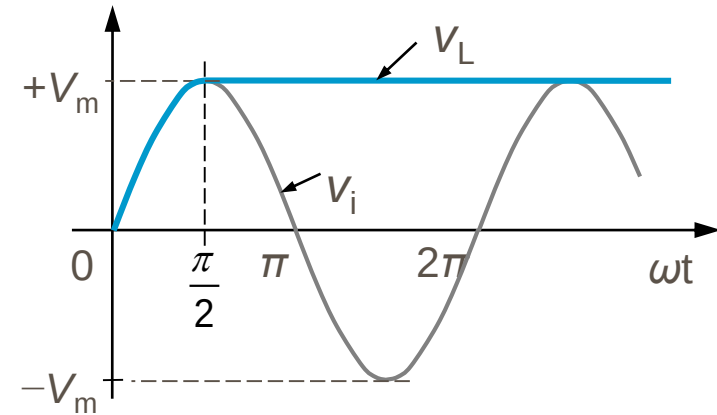
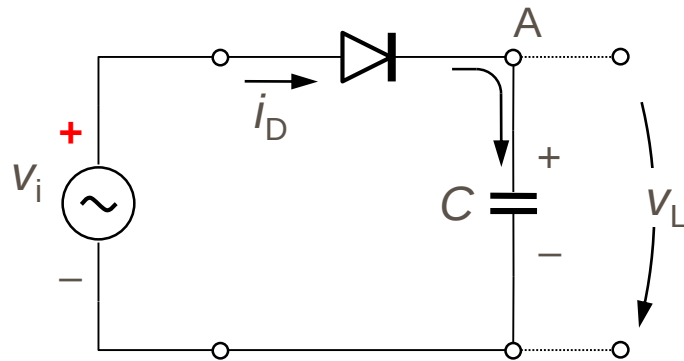
■ Comportamento do Díodo Ideal

■ Detector de Pico



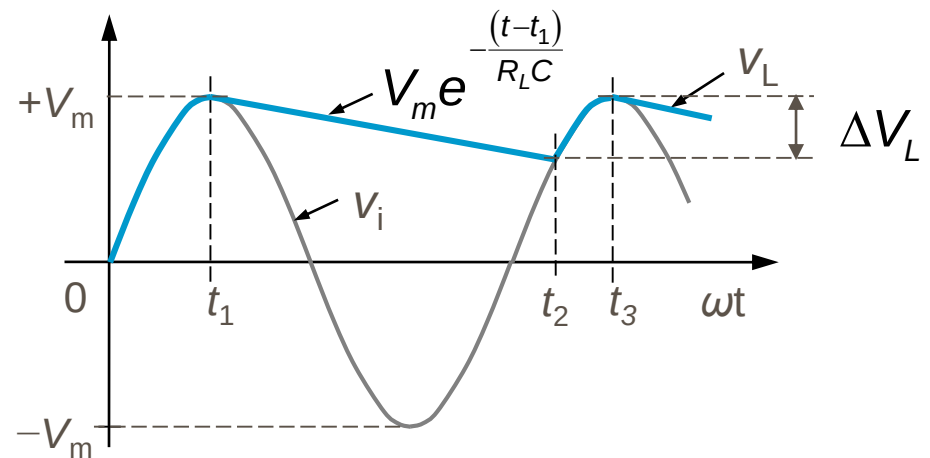
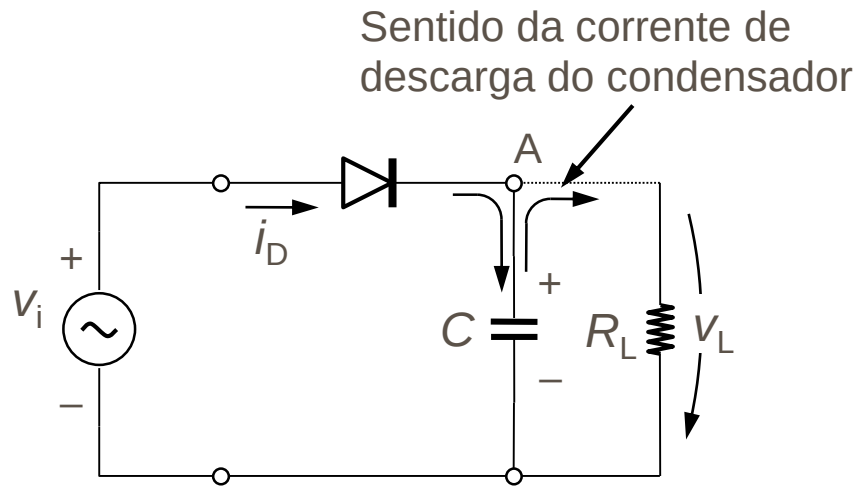
■ Comportamento do Díodo Ideal

■ Detector de Pico



■ Comportamento do Díodo Ideal

■ Detector de Pico (com carga)



Admitindo que $i_D \approx cte = \frac{V_m}{R_L}$

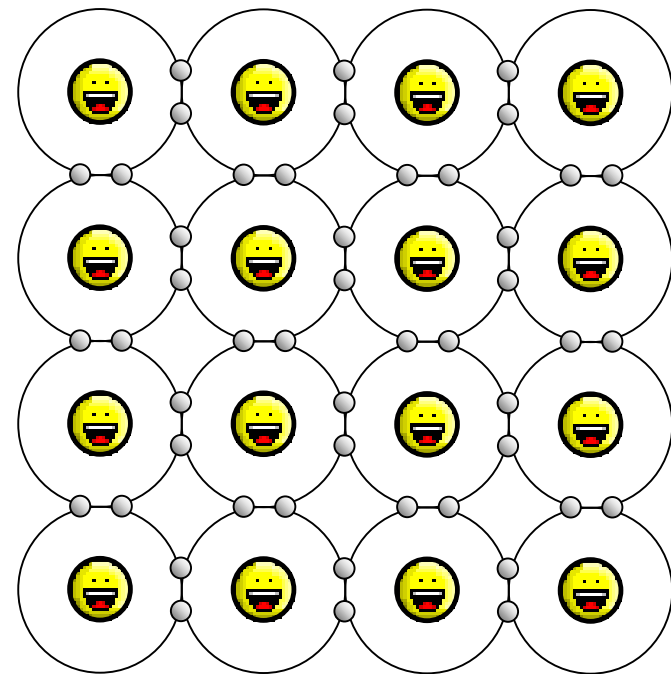
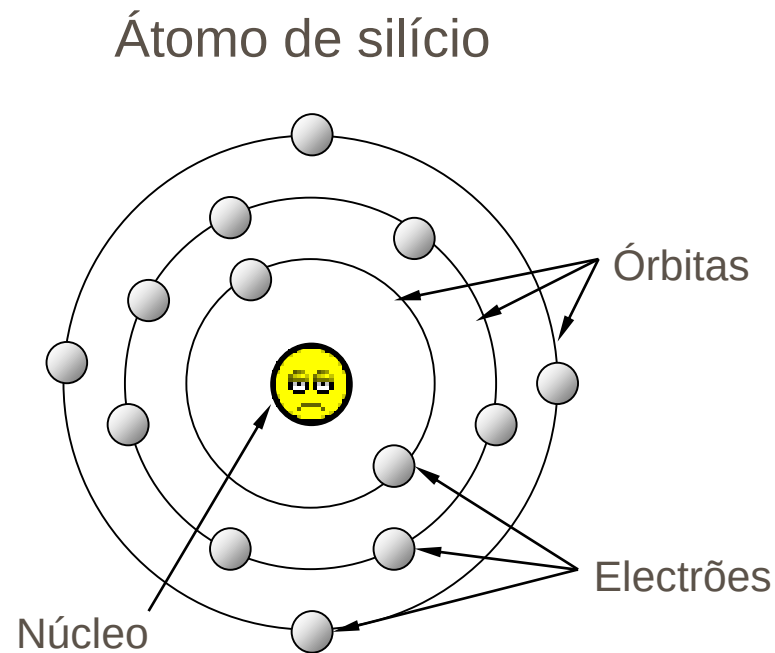
$$i_D \approx C \frac{\Delta V_L}{\Delta t} \rightarrow \Delta V_L = \frac{i_D}{C} \Delta t \approx \frac{V_m}{f \cdot R_L C}$$

O valor médio da tensão de saída é:

$$\rightarrow V_L \approx V_m - \frac{1}{2} \Delta V_L = V_m \left(1 - \frac{V_m}{2f \cdot R_L C} \right)$$

■ Princípios de física dos semicondutores

■ Funcionamento do díodo semicondutor – junção P-N



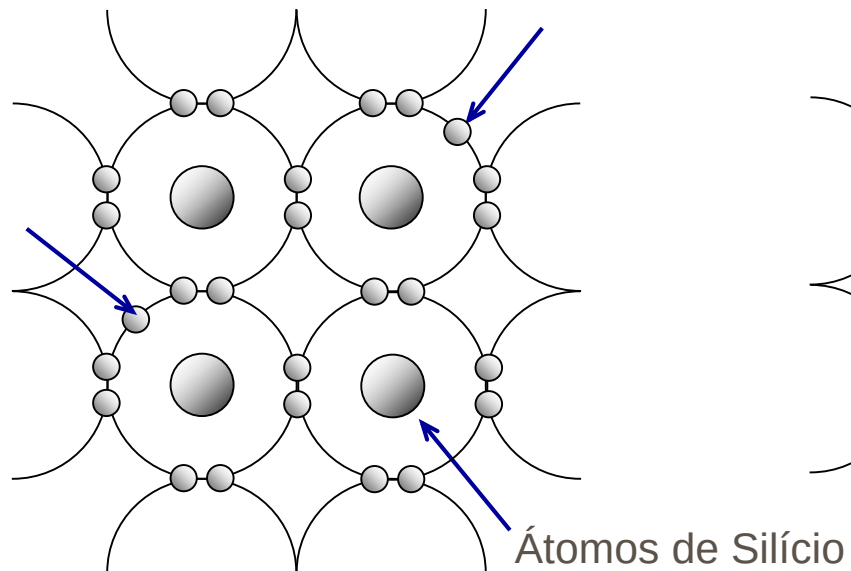
Cristal de silício (estrutura)

■ Princípios de Física dos Semicondutores

■ Funcionamento do díodo semicondutor – junção P-N

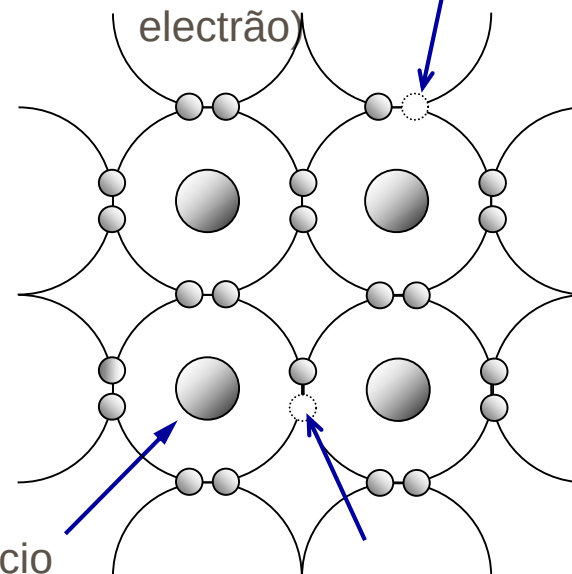
Semicondutor tipo-N

Resultado da dopagem:
electrões extra



Semicondutor tipo-P

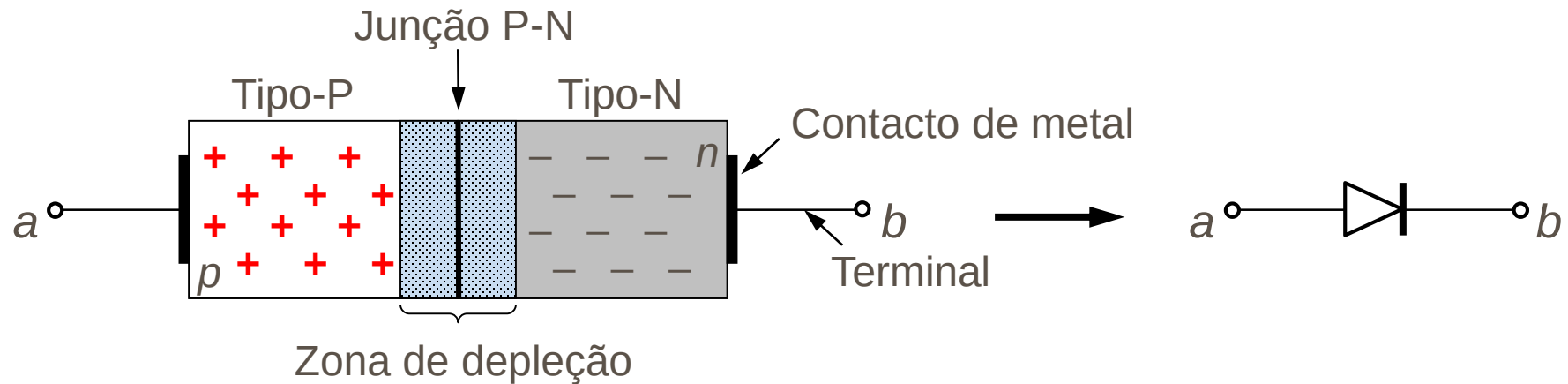
Resultado da dopagem:
lacunas (ausência de um
electrão)



Díodo Semicondutor

■ Princípios de Física dos Semicondutores

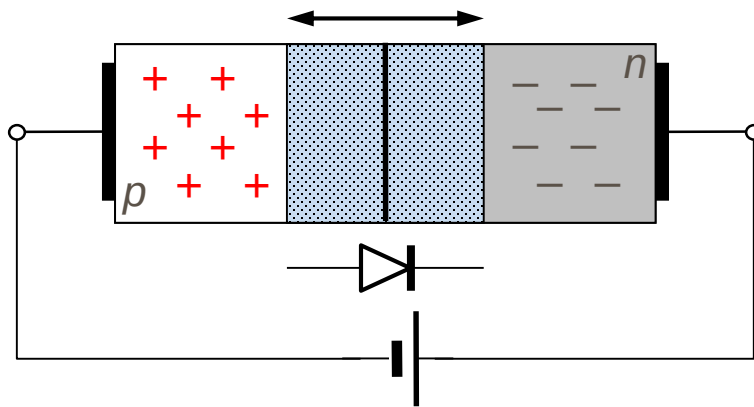
■ Funcionamento da junção P-N



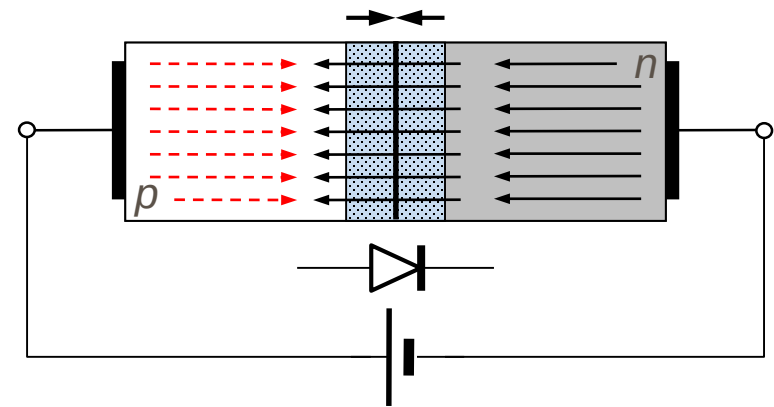
■ Princípios de Física dos Semicondutores

■ Funcionamento da junção P-N

Junção inversamente polarizada

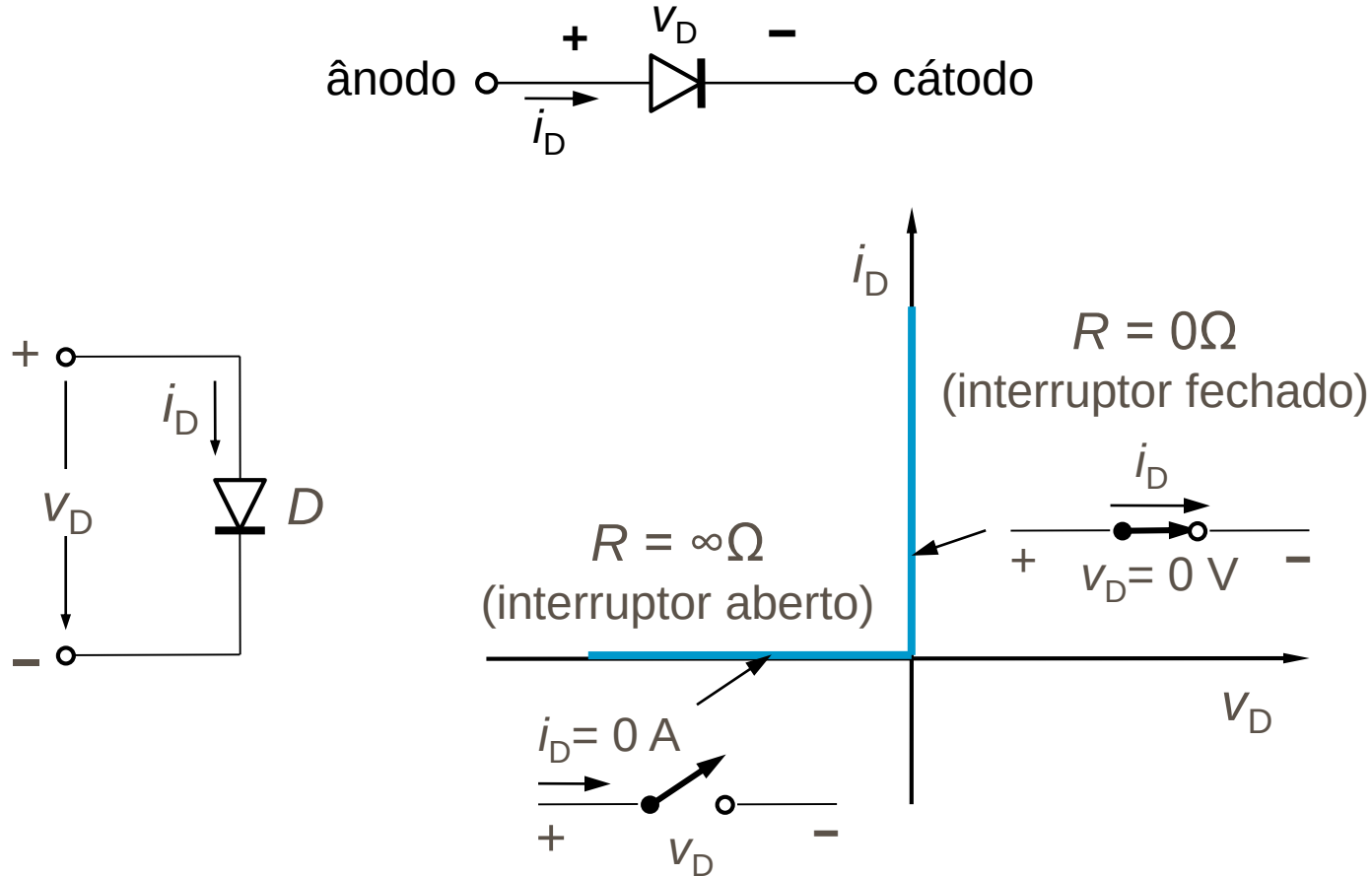


Junção directamente polarizada

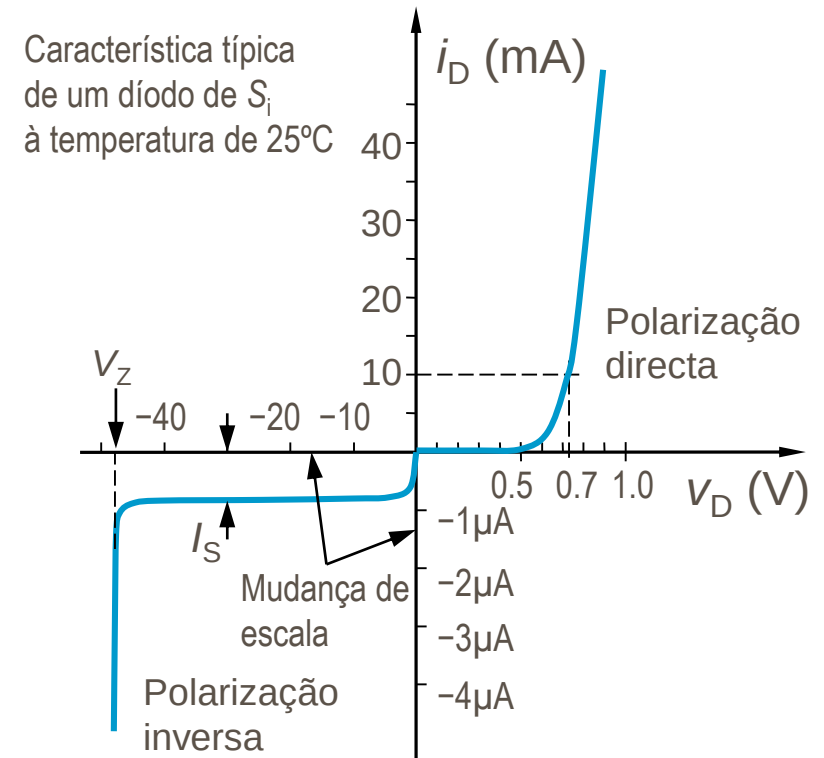
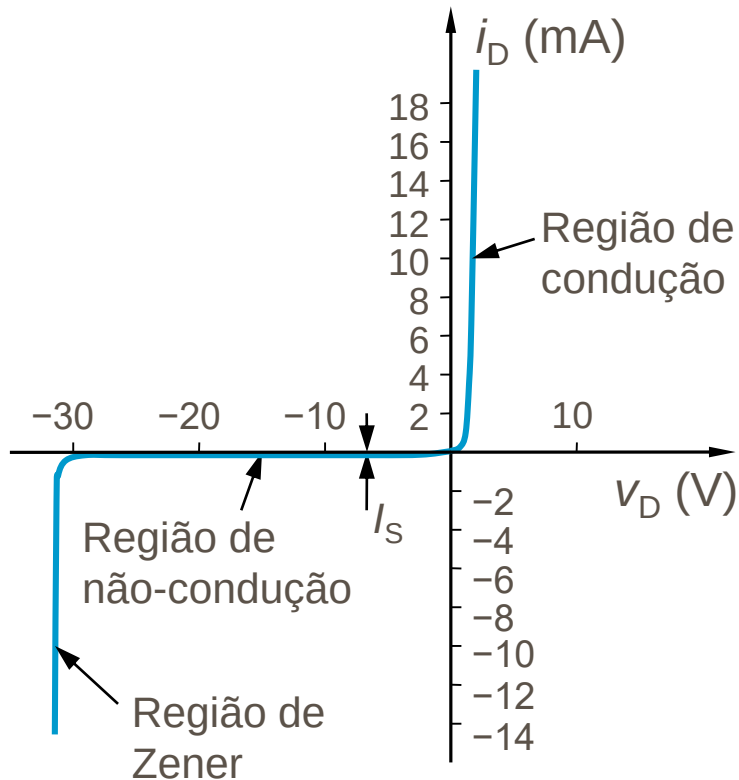


Díodo Semicondutor

■ Característica v - i do Díodo Ideal



■ Característica V-I do Díodo Semicondutor



■ Característica V-I do Díodo Semicondutor

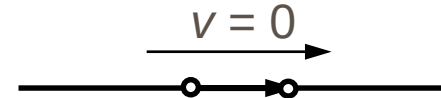
Desligado



Interruptor ideal

$$p = v \cdot i = v \cdot 0A = 0 \text{ W}$$

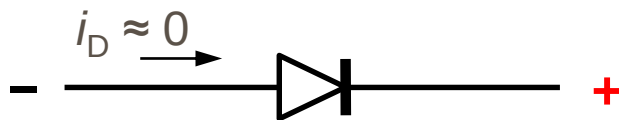
Ligado



Interruptor ideal

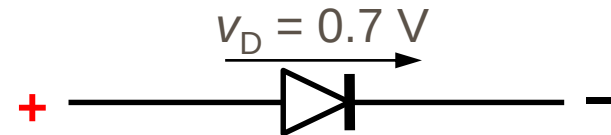
$$p = v \cdot i = 0V \cdot i = 0 \text{ W}$$

Inversamente polarizado



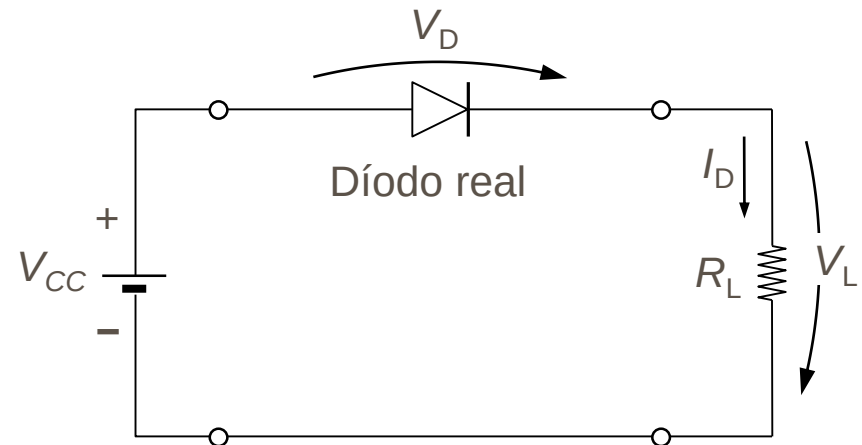
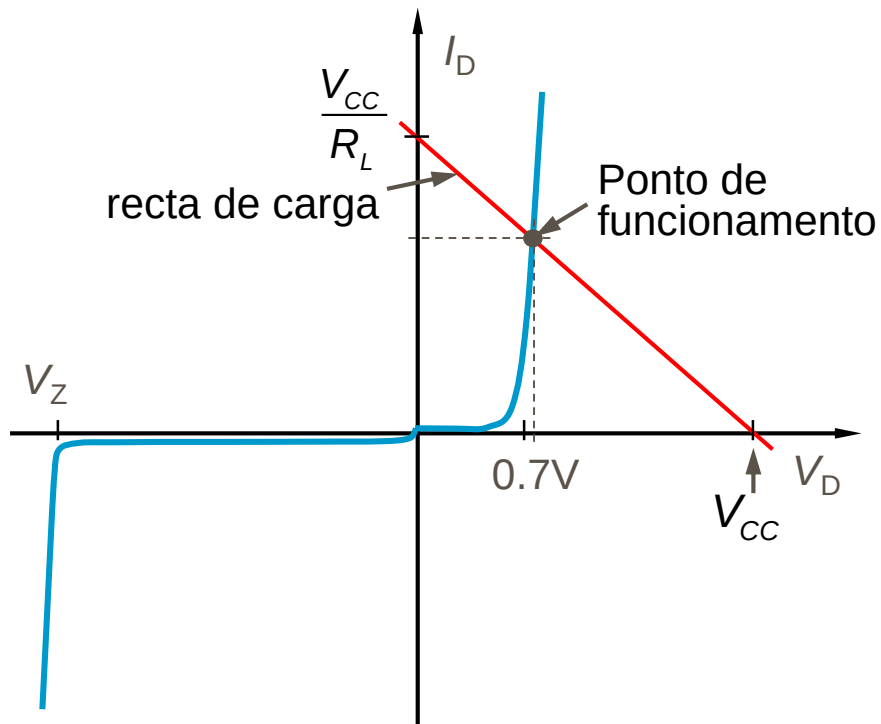
$$p_D = v_D \cdot i_D \approx v_D \cdot 0A = 0 \text{ W}$$

Directamente polarizado



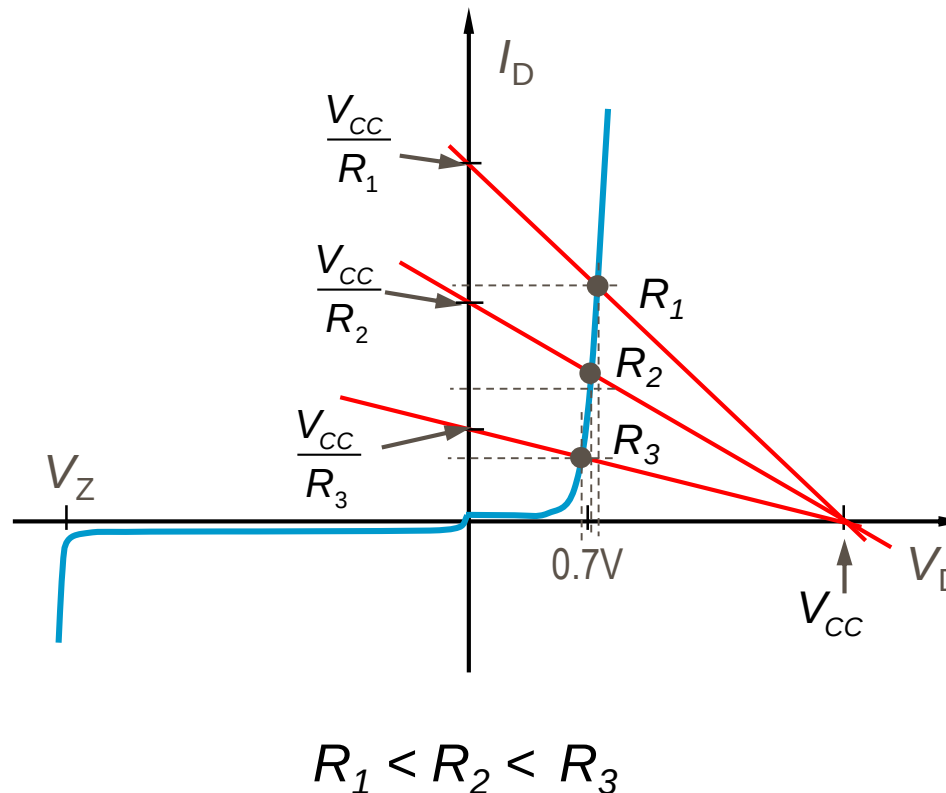
$$p_D = v_D \cdot i_D \approx 0.7V \cdot i_D$$

■ Polarização do Díodo Semicondutor



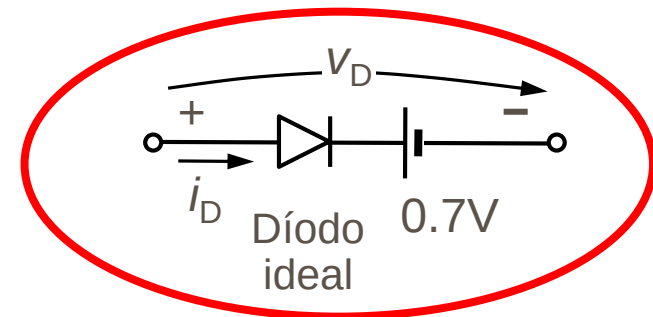
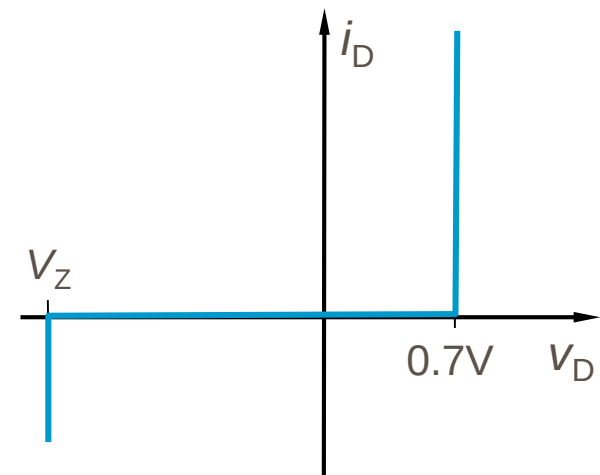
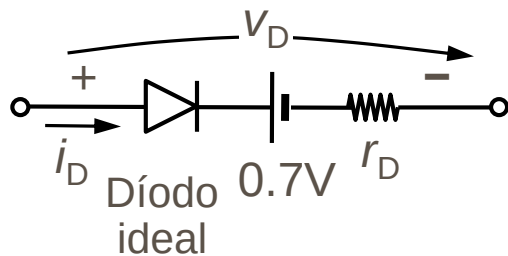
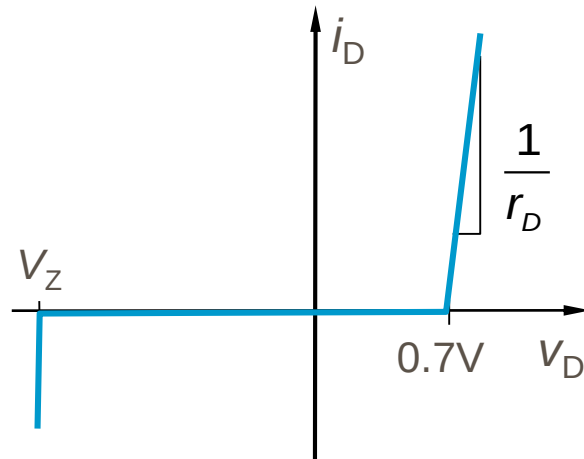
$$V_{CC} = R_L I_D + V_D \rightarrow \text{Recta de carga}$$

■ Polarização do Díodo Semicondutor



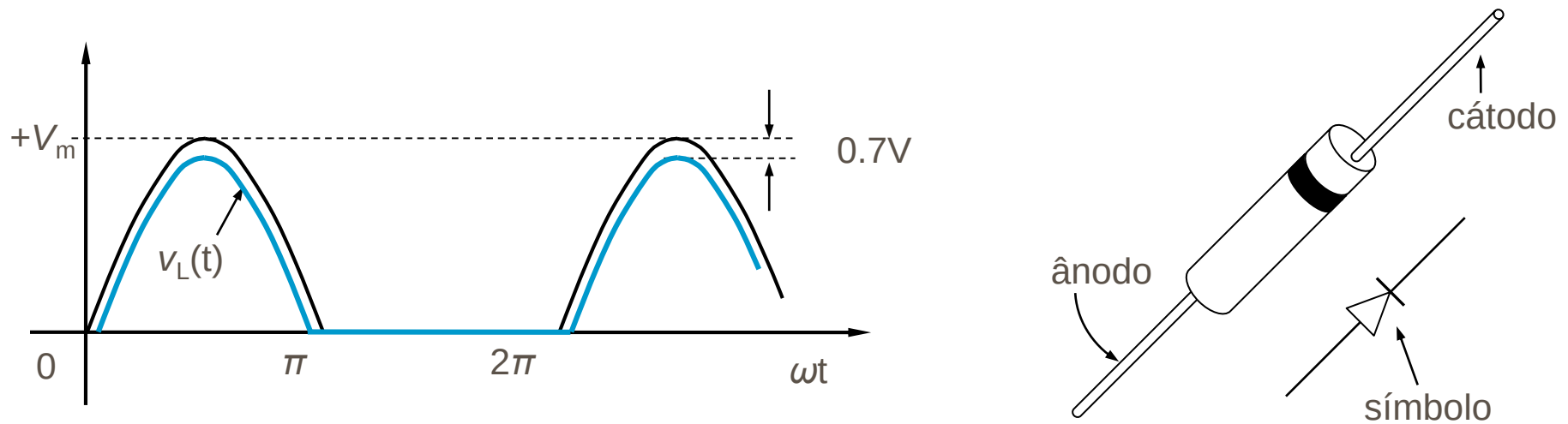
Díodo Semicondutor

■ Aproximações da Característica V - I do Díodo Semicondutor



Díodo Semicondutor

■ Comportamento do Díodo Semicondutor



Díodo Semicondutor

■ Tipos de Díodos

Pequenos Sinais



Emissor de Luz (LED)



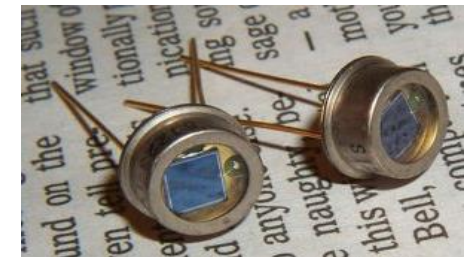
Díodo de Zener



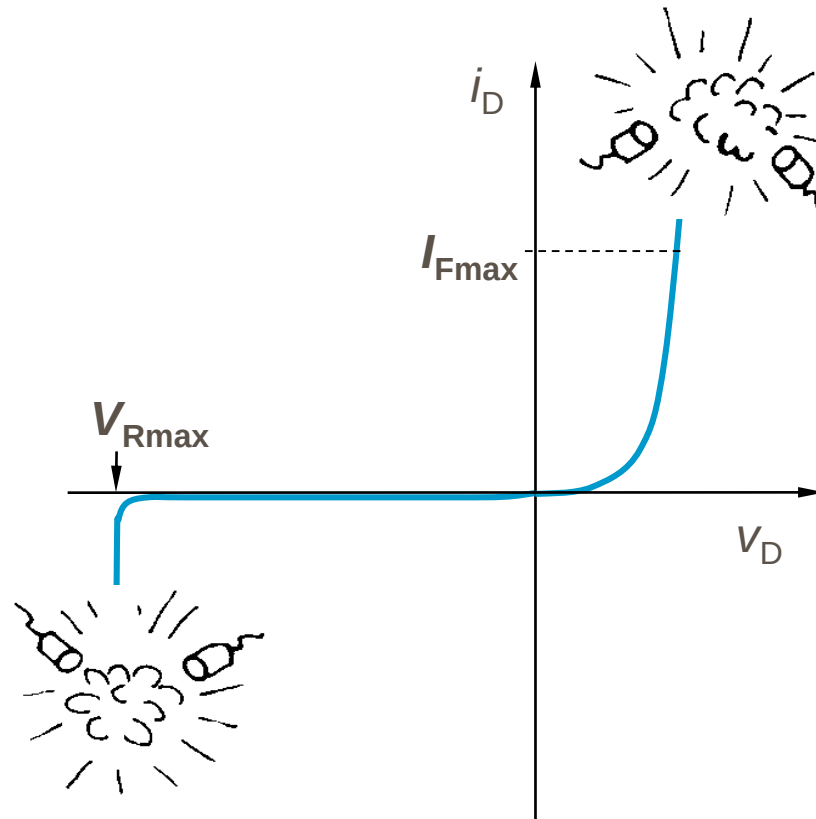
Díodo de Potência



Fotodíodo

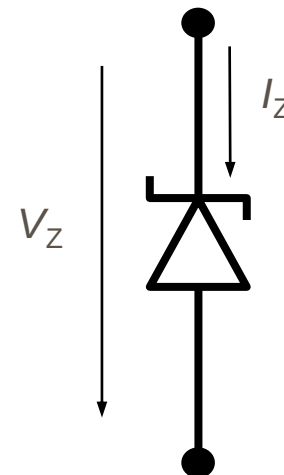
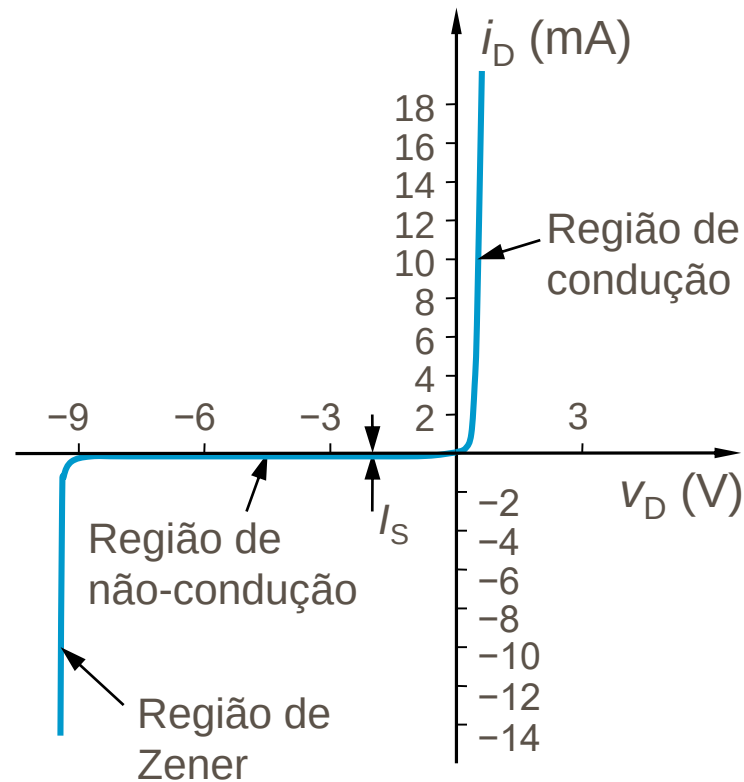


■ Limites de Operação dos Díodos



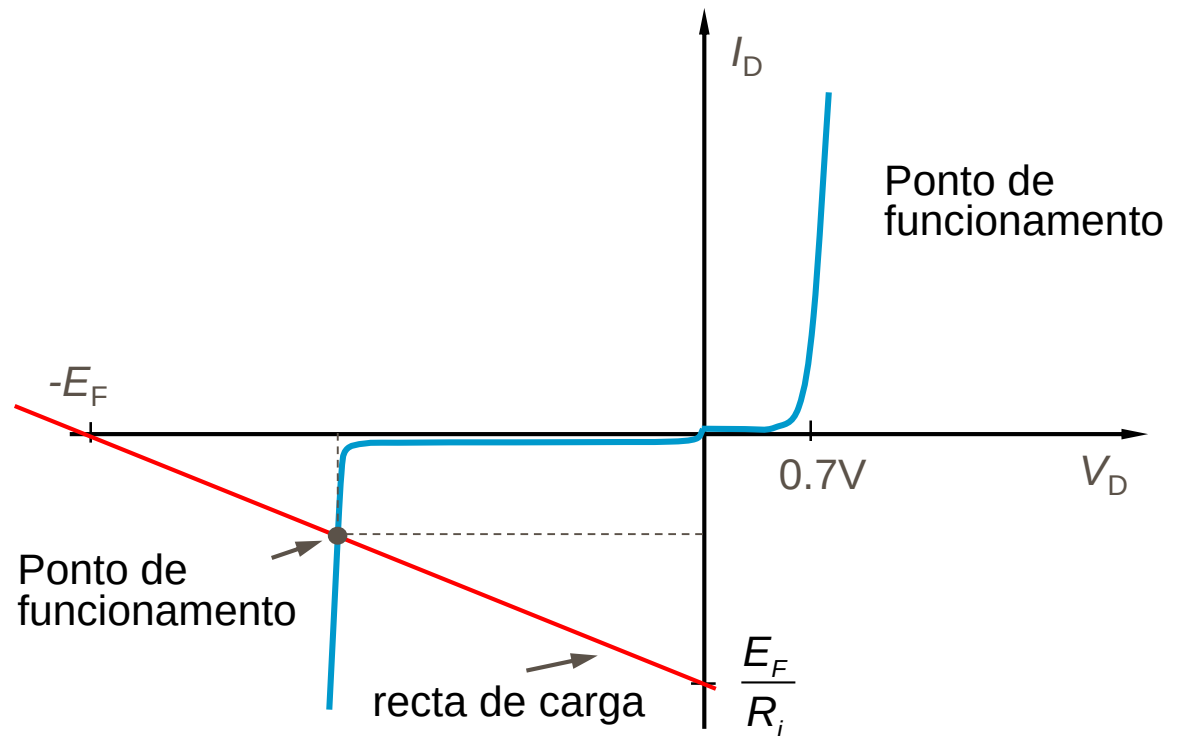
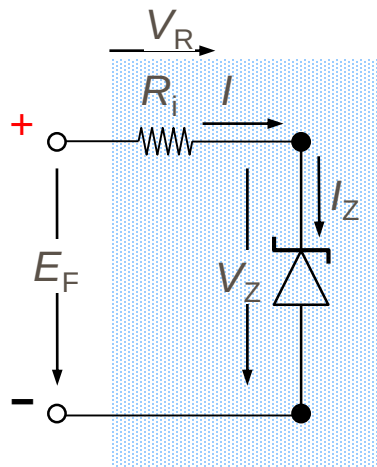
Díodo Zener

- O díodo zener é um díodo construído especialmente para trabalhar na zona da tensão de rutura.



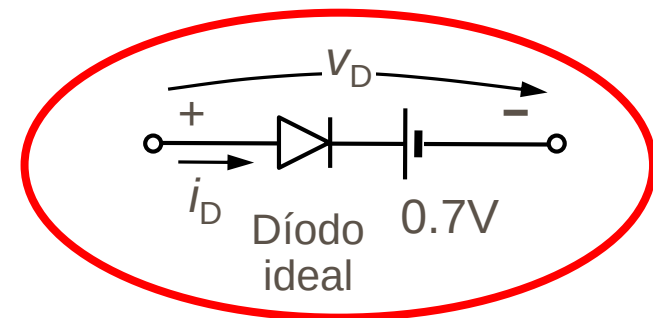
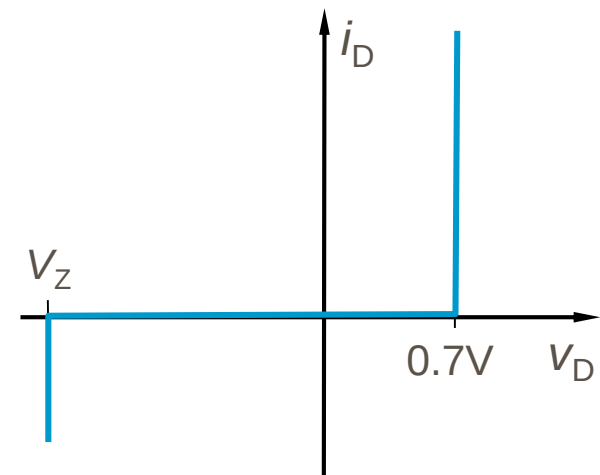
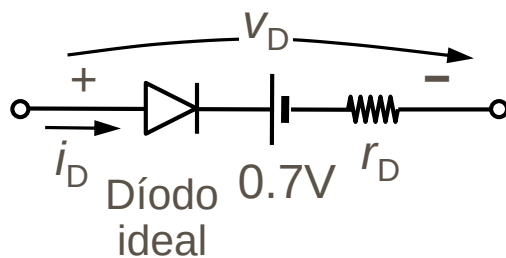
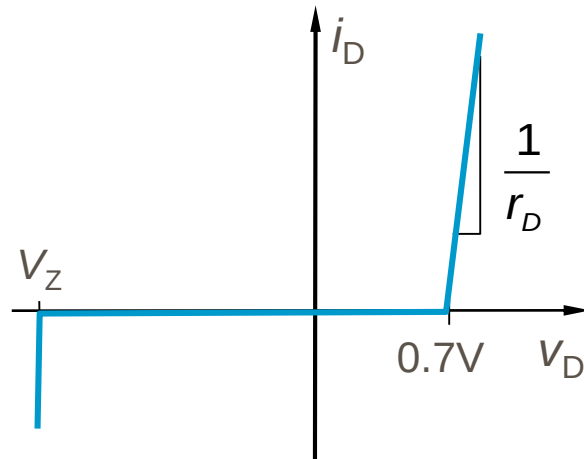
Díodo Zener

- Normalmente o díodo zener está polarizado inversamente e em série com uma resistência limitadora de corrente. Graficamente é possível obter a corrente elétrica sob o díodo zener através da reta de carga.



Díodo Semicondutor

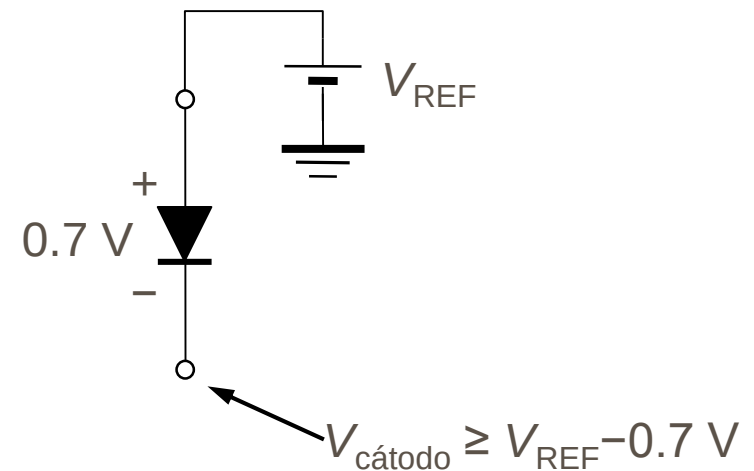
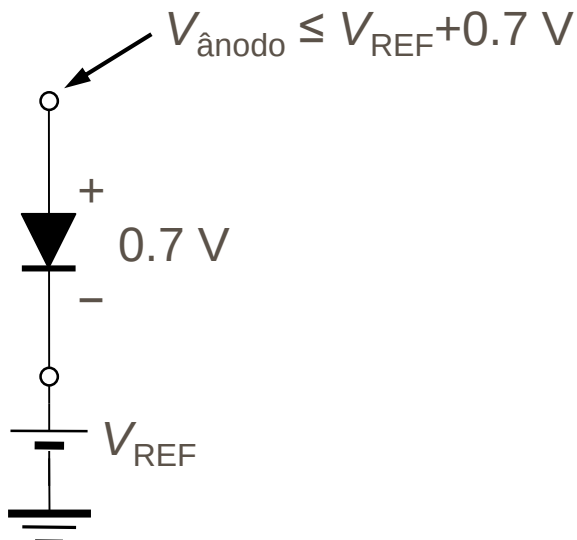
■ Aproximações da Característica V - I do Díodo Semicondutor



Díodos – Aplicações Básicas

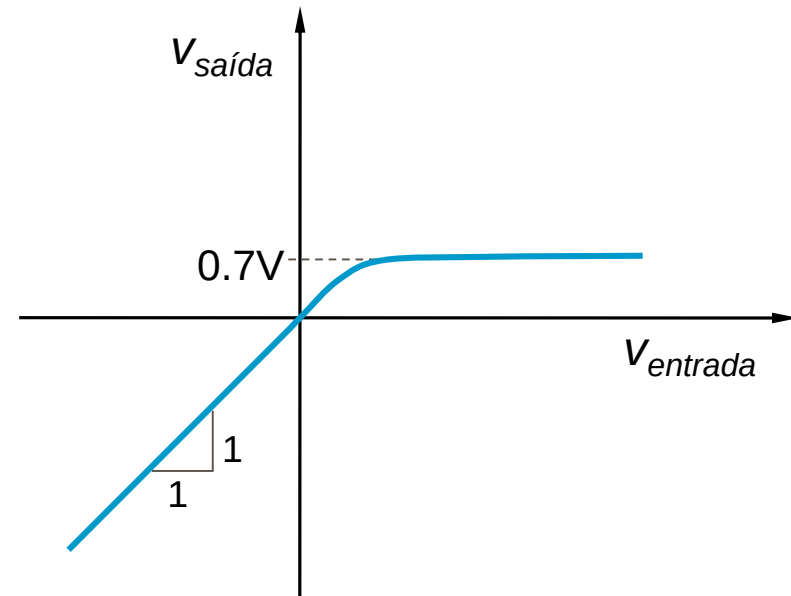
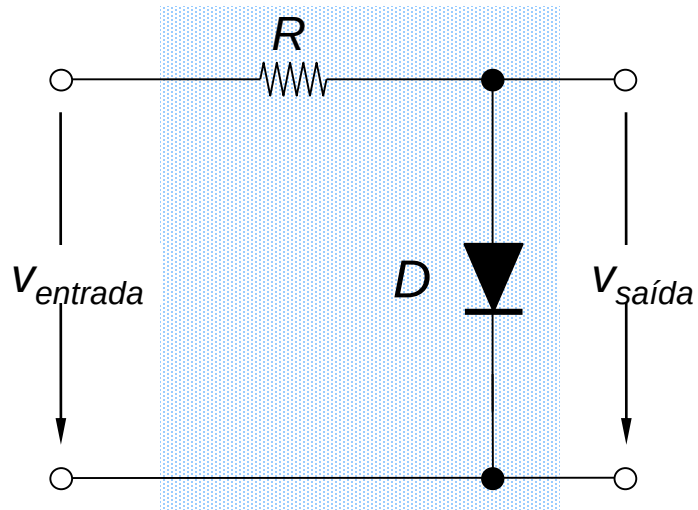
■ Circuitos Limitadores (ou *Clipping*)

- Permitem limitar a tensão em determinados pontos de um circuito
- Permitem alterar a forma de onda de um sinal



Díodos – Aplicações Básicas

■ Circuitos Limitadores



Característica de transferência

$$V_{entrada} < 0.7 V \rightarrow$$

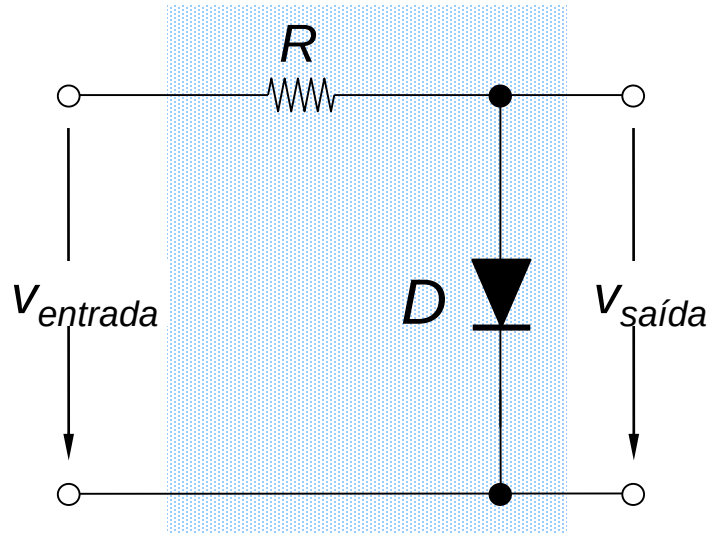
$$V_{saída} = V_{entrada}$$

$$V_{entrada} \geq 0.7 V \rightarrow$$

$$V_{saída} = 0.7 V$$

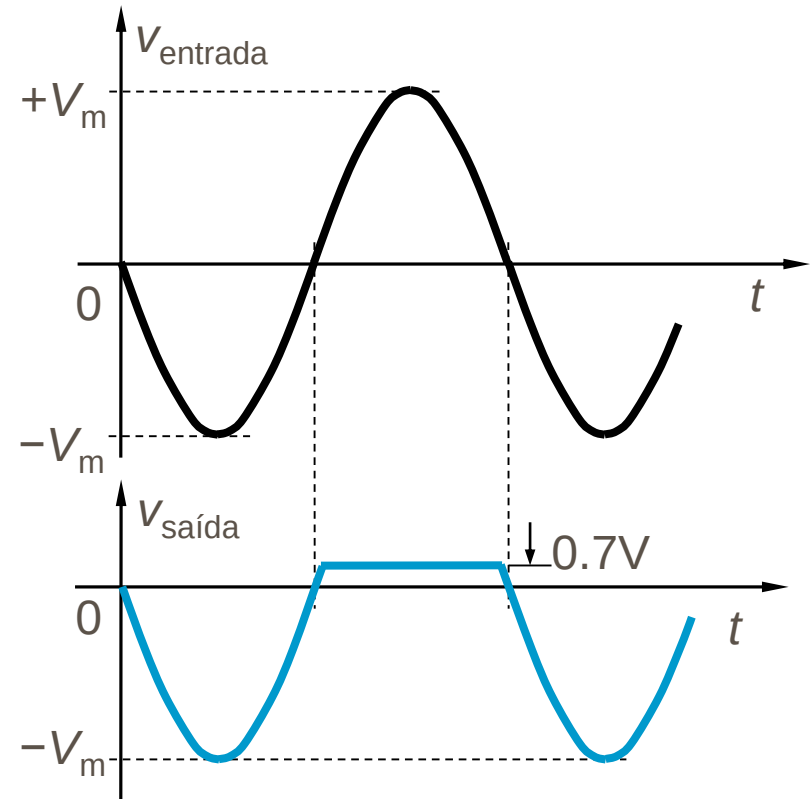
Díodos – Aplicações Básicas

■ Circuitos Limitadores



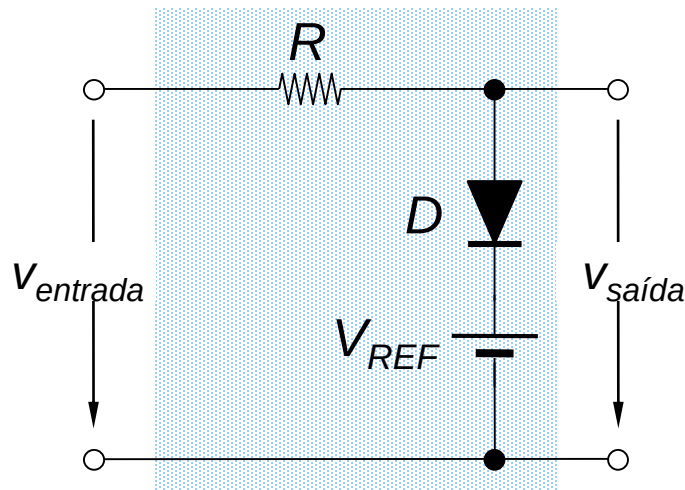
$$V_{\text{entrada}} < 0.7 \text{ V} \rightarrow V_{\text{saída}} = V_{\text{entrada}}$$

$$V_{\text{entrada}} \geq 0.7 \text{ V} \rightarrow V_{\text{saída}} = 0.7 \text{ V}$$



Díodos – Aplicações Básicas

■ Limitadores Polarizados

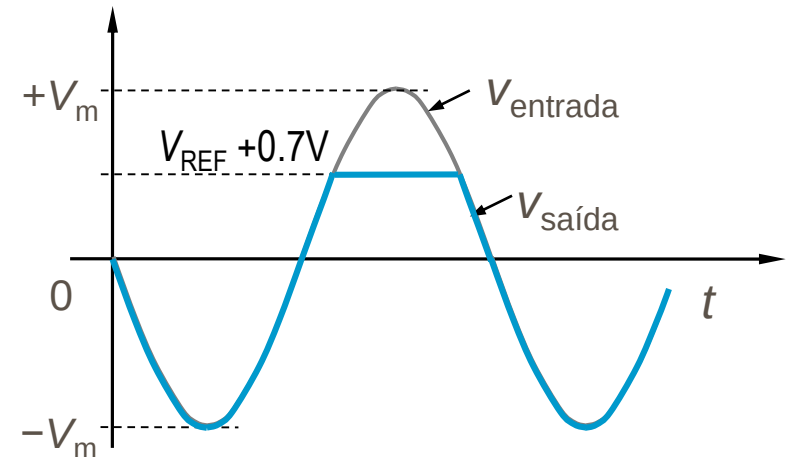
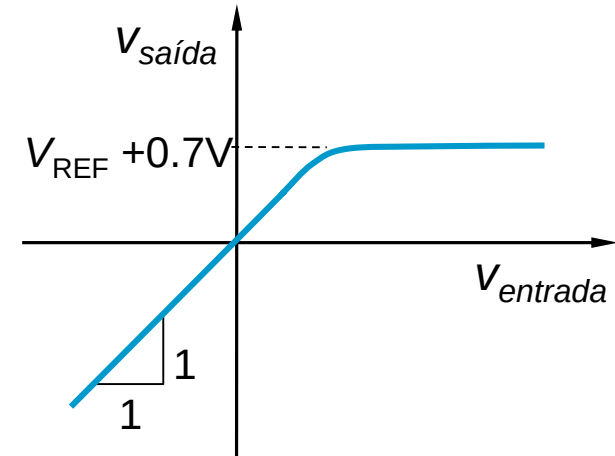


$$V_{\text{entrada}} < V_{\text{REF}} + 0.7 \text{ V} \rightarrow$$

$$V_{\text{entrada}} \geq V_{\text{REF}} + 0.7 \text{ V} \rightarrow$$

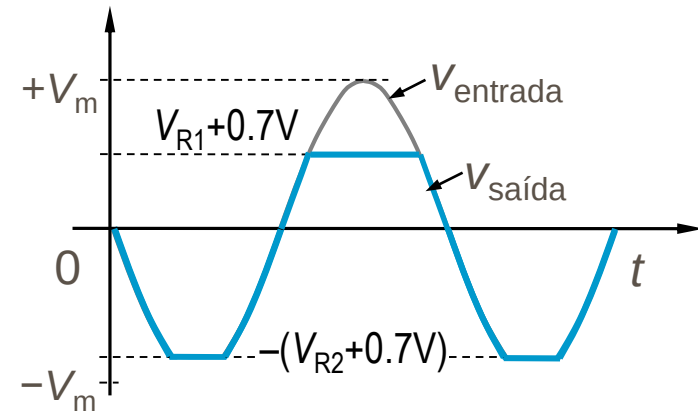
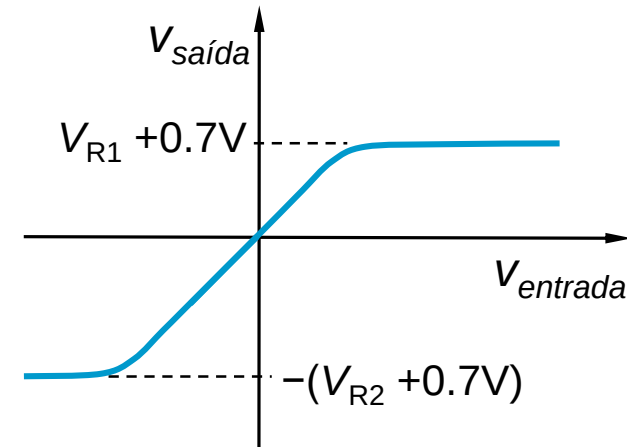
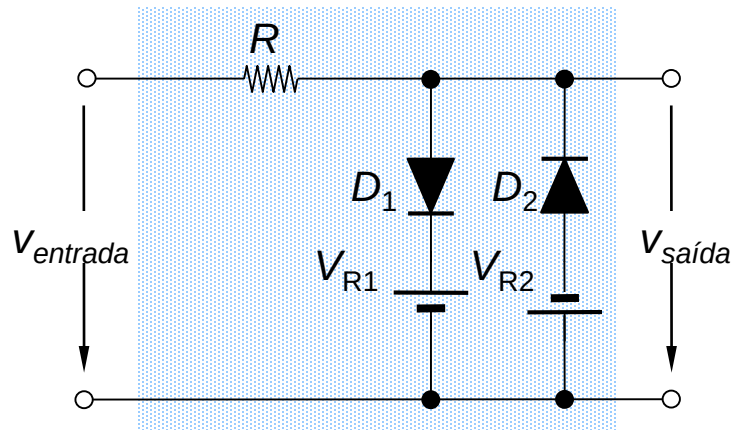
$$V_{\text{saída}} = V_{\text{entrada}}$$

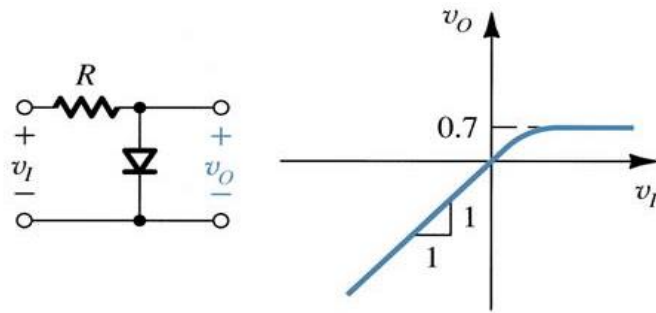
$$V_{\text{saída}} = V_{\text{REF}} + 0.7 \text{ V}$$



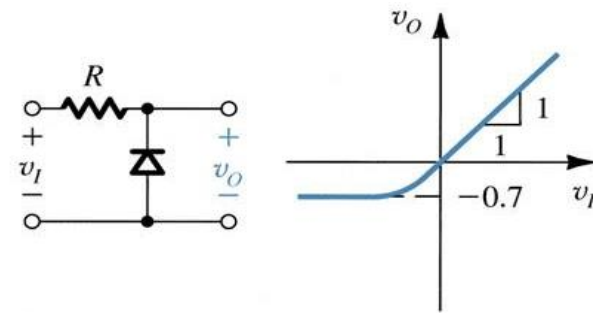
Díodos – Aplicações Básicas

■ Limitação a 2 níveis

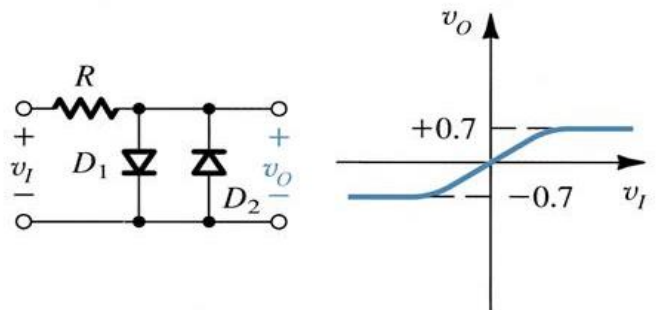




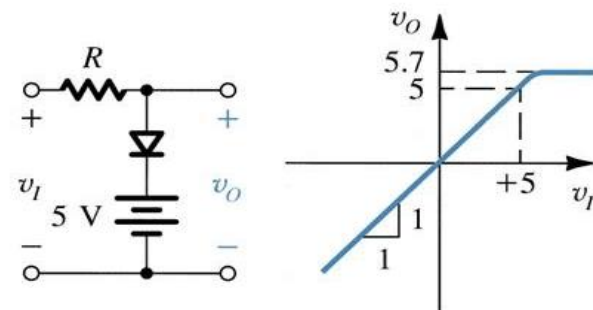
(a)



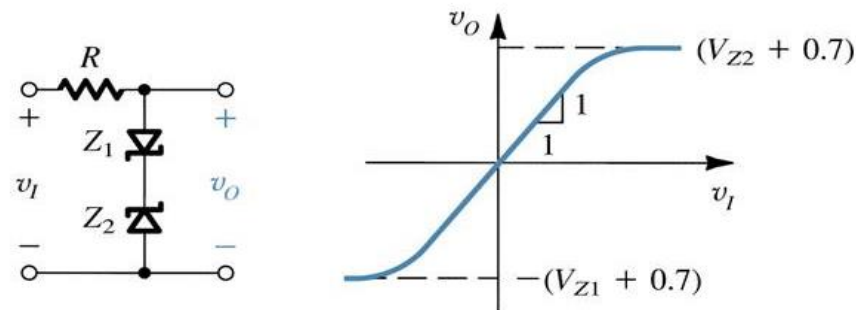
(b)



(c)



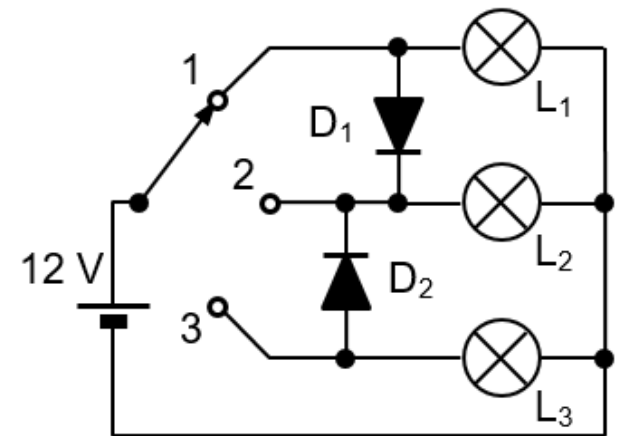
(d)



(e)

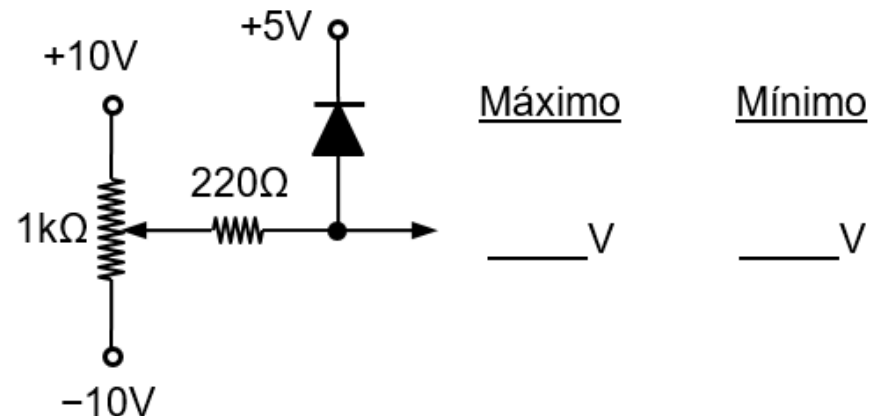
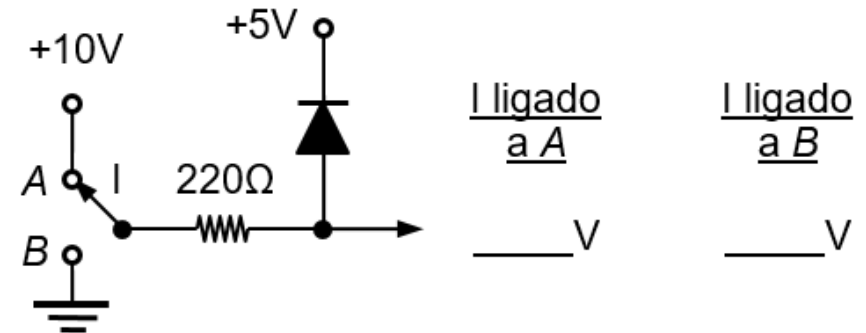
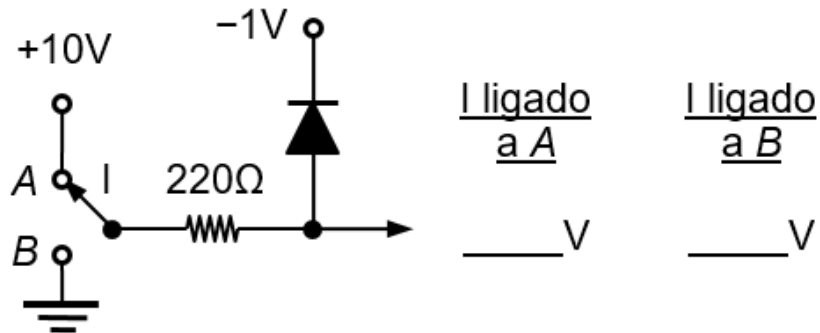
Exercícios

- 33 No circuito da Figura 28, L_1 , L_2 e L_3 são lâmpadas de 12V. Qual (ou quais) da(s) lâmpada(s) acende quando o comutador está na posição 1, 2, e 3?



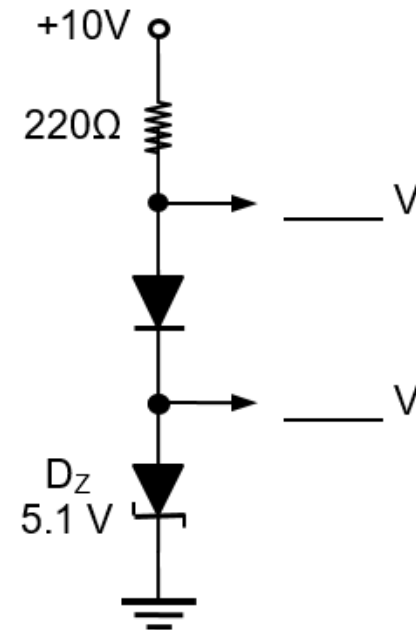
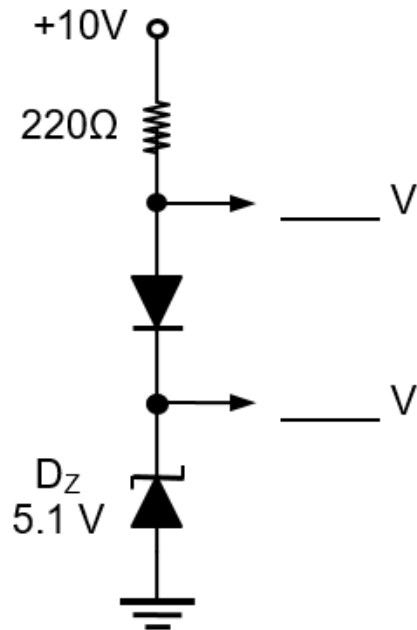
Exercícios

- 31 Calcule, para os circuitos seguintes, o potencial nos pontos indicados (considere a queda de tensão de condução direta nos díodos igual a 0,7 V).



Exercícios

- 36 Calcule, para o circuito da Figura 31, o potencial nos pontos indicados (considere a queda de tensão de condução directa nos díodos igual a 0,7 V):



Fim da Aula 2